

SOLID GAS MASS TRANSFER

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Objective:

The aim of this experiment is to determine the mass transfer coefficient for vaporization of naphthalene balls in air by investigating the solid-gas mass transfer process. This study aims to analyze the rate of sublimation and understand the principles of diffusion in a controlled experimental setup.

Theory and Background:

Solid-gas mass transfer occurs when a substance moves from a solid surface into a surrounding gas due to a concentration gradient. This transfer is usually regulated by diffusion, in which molecules travel from higher concentration to lower concentration, influenced by a concentration gradient. This process follows Fick's Law of Diffusion, which states that the mass flux is proportional to the concentration gradient and the diffusion coefficient.

Fick's law can be mathematically written as:

$$J = - D A \frac{dC}{dx}$$

Where:

- $J = \text{Flux of naphthalene molecules} \left(\frac{\text{mol}}{\text{m}^2 \text{ s}} \right)$
- $D = \text{Diffusion coefficient of naphthalene in air} \left(\frac{\text{m}^2}{\text{s}} \right)$
- $A = \text{Surface area of the naphthalene balls} (\text{m}^2)$
- $\frac{dC}{dx} = \text{Concentration gradient of naphthalene molecules}$

In the case of naphthalene balls, sublimation takes place, where solid naphthalene directly transitions into its gaseous state. The higher concentration of naphthalene molecules at the surface creates a concentration gradient, driving mass transfer into the air. This process is influenced by several factors, including surface area, air velocity, temperature, and pressure.

The equation for mass transfer to gas phase can be written as:

$$N_A = k_G (P_A^1 - P_A^2)$$

Where:

- N_A is rate of mole transfer per unit area
- k_G is the mass transfer coefficient for gas phase
- P_A^1 is partial pressure of given substance at point 1
- P_A^2 is partial pressure of given substance at point 2

Sublimation is a direct phase transition from solid to gas without passing through the liquid phase. The sublimation rate is determined by the difference in partial pressure of naphthalene between the solid surface and the bulk air.

Naphthalene is a volatile organic compound that easily sublimates into the air under normal conditions, emitting aromatic vapors. Due to this property, naphthalene balls are commonly used in experiments to study solid-gas mass transfer.

The experiment assumes steady-state conditions, where sublimation occurs uniformly over time due to continuous airflow and stable temperature. The study of this mass transfer mechanism is essential for applications such as drying operations, sublimation-based separation processes, and the environmental dispersion of volatile compounds.



Figure 0: Figure showing the experimental set-up

Apparatus:

1. Naphthalene Balls
2. String (to hold 10 naphthalene balls)
3. Vertical tube
4. Dryer
5. Vernier calipers

Procedure:

1. Weigh all the naphthalene balls together and note down the initial total weight.
2. Use Vernier Calipers to measure and record the height and diameter of each naphthalene ball individually.
3. Suspend the naphthalene balls in the vertical tube and turn on the dryer.
4. Allow the dryer to run for 30 minutes, then remove the balls from the tube.
5. Weigh the naphthalene balls again and record the new total weight.
6. Measure the height and diameter of each ball again to note the changes.
7. Repeat steps 3 to 6 for a total of three trials, ensuring consistent drying intervals.
8. Record all the weight and size changes after each trial for analysis.

Precautions:

1. Wear gloves at all times when handling naphthalene balls.
2. Make sure the naphthalene balls and the string of naphthalene balls don't fall in the drying section while attaching the thread on the top of the apparatus.
3. Mark the measurement points on the naphthalene balls, as even slight variations in position can affect accuracy.

Known parameters:

- *Molecular weight of Naphthalene:* 128.1705 g
- *Saturated Pressure of Naphthalene at 25°C* = 11.714 Pa

Formulae:

- $\Delta M = M_1 - M_2$, where:
 - $M_1 = \text{mass of Naphthalene balls at time } t = t_1$
 - $M_2 = \text{mass of Naphthalene balls at time } t = t_2$

- $\Delta N = \frac{\Delta M}{\text{Molecular weight of Naphthalene}}$
- $\text{Total Surface area} = 2\pi r (h + r) * 9$ (total for 9 balls)
 - There were 10 balls initially but one ball was broken so we had 9 balls for the experiment.
 - r is the average radius of 9 naphthalene balls
 - h is the average height of 9 naphthalene balls
- $\text{Flux} = \frac{\Delta N}{\text{Time} * \text{Total Surface Area}}$
- $K_G = \frac{\text{Flux}}{\text{Saturated Pressure of Naphthalene at } 25^\circ\text{C}}$

Readings:

TABLE 1: TABLE CONTAINING EXPERIMENTAL DATA OF MASS OF NAPHTHALENE BALLS AND TIME

Time (min)	Mass (g)
0	46.04
30	44.6
60	43.44
90	42.25
120	40.55

TABLE 2: TABLE CONTAINING EXPERIMENTAL DATA OF HEIGHT OF NAPHTHALENE BALLS WITH TIME

Naphthalene ball number (from top to bottom)	Height of Naphthalene ball at t = 0 min (mm)	Height of Naphthalene ball at t = 30 min (mm)	Height of Naphthalene ball at t = 60 min (mm)	Height of Naphthalene ball at t = 90 min (mm)	Height of Naphthalene ball at t = 120 min (mm)
1	14.74	14.33	14.21	14.11	13.91
2	14.61	14.24	14.13	14.05	13.93
3	14.46	14.24	14.02	13.95	13.84
4	14.71	14.44	14.21	14.08	13.92
5	14.66	14.51	14.25	14.12	13.97
6	14.65	14.35	13.92	13.91	13.82
7	14.51	14.44	13.9	13.68	13.56
8	14.86	14.52	14.18	13.99	13.79

9	14.58	14.4	14.21	13.92	13.85
Average height (mm)	14.64	14.38	14.11	13.97	13.84

TABLE 3: TABLE CONTAINING EXPERIMENTAL DATA OF RADIUS OF NAPHTHALENE BALLS WITH TIME

Naphthalene ball number (from top to bottom)	Diameter of Naphthalene ball at t = 0 min (mm)	Diameter of Naphthalene ball at t = 30 min (mm)	Diameter of Naphthalene ball at t = 60 min (mm)	Diameter of Naphthalene ball at t = 90 min (mm)	Diameter of Naphthalene ball at t = 120 min (mm)
1	21.36	21.16	20.94	20.81	20.61
2	21.2	21.09	20.73	20.69	20.52
3	21.31	21.12	20.67	20.46	20.09
4	21.32	21.13	20.68	20.57	20.38
5	21.27	21.05	20.7	20.51	20.24
6	21.35	21.04	20.75	20.69	20.49
7	21.23	21	20.62	20.39	20.01
8	21.38	21.03	20.95	20.88	20.64
9	21.32	21.01	20.41	20.21	19.9
Average Diameter (mm)	21.30444444	21.07	20.71666667	20.57888889	20.32

TABLE 4: TABLE CONTAINING CALCULATED DATA

Time (min)	Average Height of ball (mm)	Average diameter of ball (mm)	Average Surface area (mm ²)	Average Surface area (m ²)	Weight loss (Δ M) (g)	Amount of naphthalene lost (Δ N) (mol)	Flux $\left(\frac{\text{mol}}{\text{m}^2 \text{ s}}\right)$	Mass transfer Coefficient $\left(\frac{\text{mol}}{\text{Pa m}^2 \text{ s}}\right)$
0	14.64	21.30	15228.86	1.52E-02	0	0	0	0
30	14.39	21.07	14838.65	1.48E-02	1.44	0.01123503458	4.21E-04	0.00003590841836
60	14.11	20.72	14327.66	1.43E-02	1.16	0.009050444525	3.51E-04	0.0000299578667
90	13.98	20.58	14113.48	1.41E-02	1.19	0.009284507746	3.65E-04	0.00003119903256
120	13.84	20.43	13886.89	1.39E-02	1.7	0.01326358249	5.31E-04	0.00004529726793

Average mass transfer Coefficient is $2.847 \times 10^{-5} \left(\frac{\text{mol}}{\text{Pa m}^2 \text{s}} \right)$

Results and discussion:

Mass transfer Coefficient (mol/Pa.m².s) vs. Time (min)

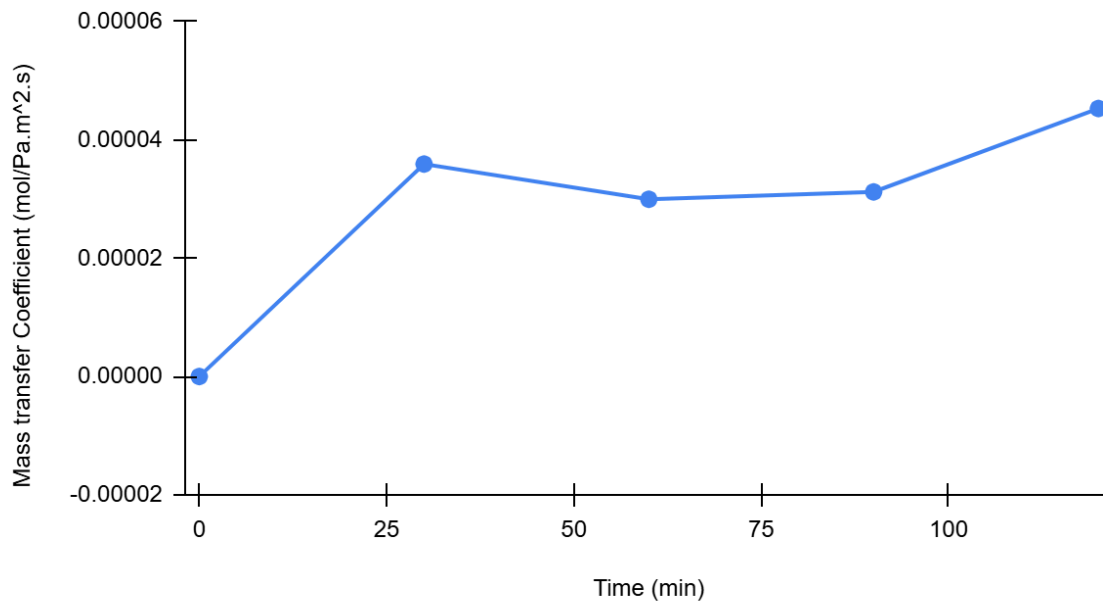


Figure 1: Figure showing the plot between Mass transfer coefficient and time

Mass of Naphthalene balls (g) vs. Time (min)

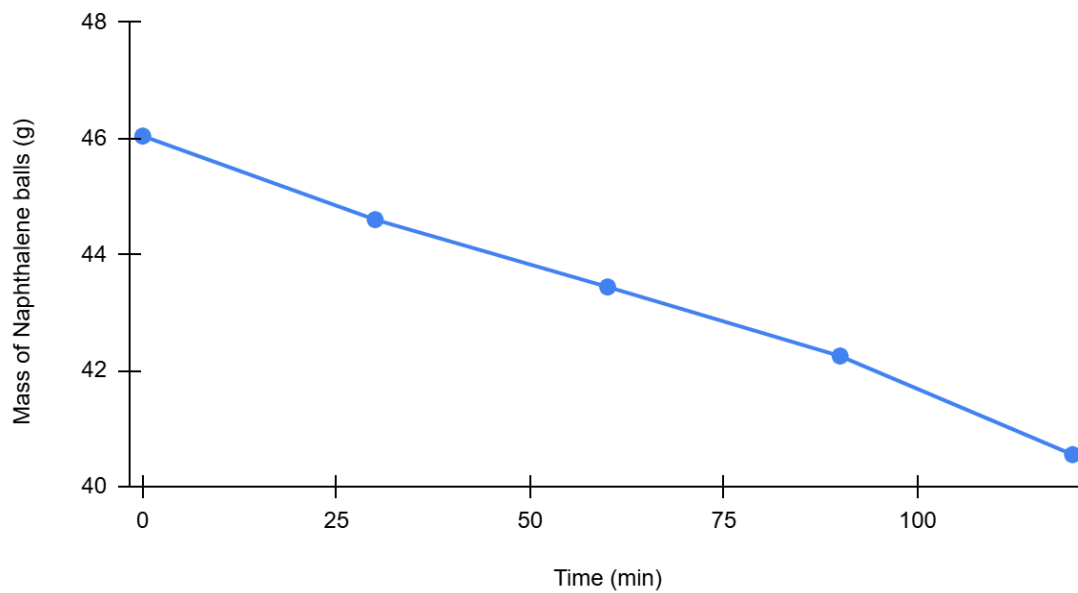


Figure 2: Figure showing the plot between Mass of naphthalene balls and time

Average Surface area (m²) vs time (min)

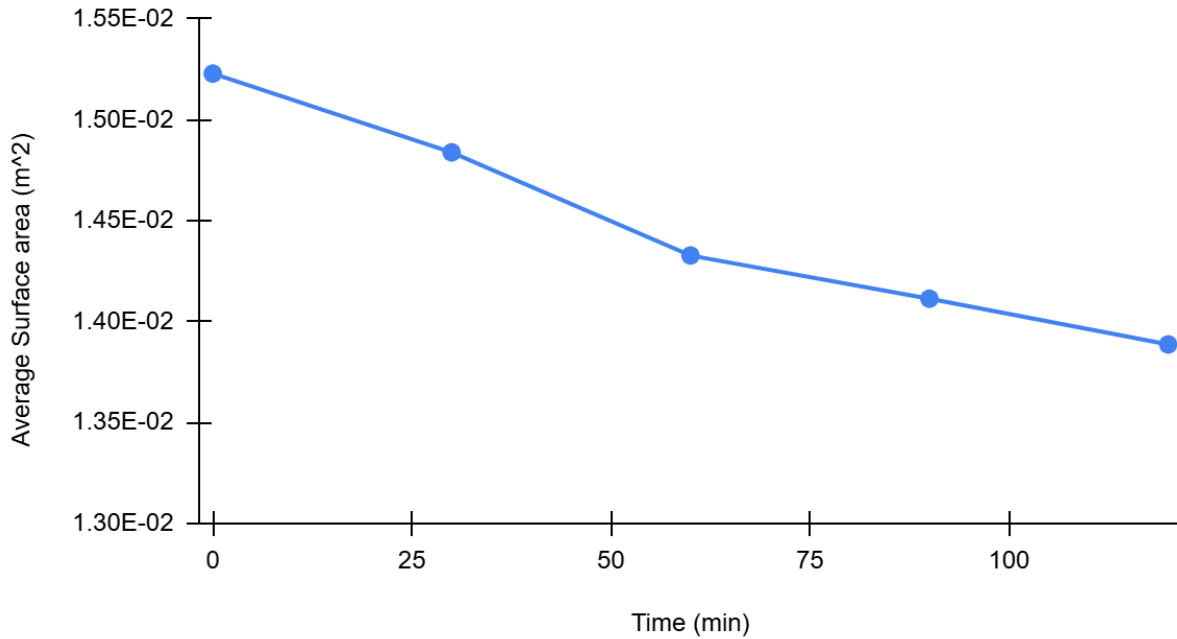


Figure 3: Figure showing the plot between Average surface area of naphthalene balls and time

From figure 2 and 3 we can see that weight and total surface area of the Naphthalene balls decreases over time as the air is blown over them. The decreasing surface area trend with the decrease in weight of the naphthalene balls confirmed that the sublimation has occurred. The weight lost by naphthalene balls can be directly attributed to the phase change of naphthalene from solid to gas, driven by the concentration gradient between the surface of the balls and the surrounding air.

From Figure 1, we can see that the mass transfer coefficient initially increases, indicating that as sublimation begins, the available surface area is relatively high, and the concentration gradient drives a faster transfer. As time progresses, the coefficient slightly decreases and stabilizes, likely due to the reduction in surface area as the naphthalene balls shrink, lowering the effective mass transfer rate. Additionally, possible saturation effects in the surrounding air or minor variations in airflow could contribute to this trend. The final increase in the mass transfer coefficient might be due to changes in air turbulence or improved exposure to fresh sublimating surfaces as the balls shrink further, enhancing the local mass transfer rate. These observations validate that mass transfer occurs through diffusion at the interface, driven by the concentration gradient between the solid naphthalene and the surrounding air. The steady reduction in weight and dimensions confirms that sublimation is uniform and primarily occurs from the surface.

Conclusion:

From this experiment we can conclude that diffusion occurs as a substance moves from a region of higher concentration to a lower one, as demonstrated in this experiment by the weight loss and size reduction of the naphthalene balls. This process facilitated the evaluation of the diffusion coefficient of naphthalene in air, which quantifies the rate at which naphthalene disperses into the surrounding medium. The diffusion coefficient was determined based on the measured weight loss over time and the established concentration gradient between the balls and the surrounding air.

The mass transfer coefficient, experimentally determined as $2.847 \times 10^{-5} \left(\frac{\text{mol}}{\text{Pa m}^2 \text{s}} \right)$, varied over time, deviating from linearity. This variation was likely influenced by changes in surface area, airflow velocity, and vapor concentration. The results highlighted the principles of diffusion and convective mass transfer, demonstrating how naphthalene sublimates due to the concentration gradient. The decrease in weight and surface dimensions of the balls further confirmed that mass transfer occurred from the solid surface to the gaseous phase.

These findings validated the application of Fick's Law of Diffusion and provided insights into how diffusion coefficients can be determined experimentally. The experiment also illustrated the relationship between surface area reduction and mass transfer rates, emphasizing its relevance in real-world applications such as drying, sublimation, and other gas-solid mass transfer operations.

Sources of errors:

- The diameter may not be measured at the exact same point every time.
- Naphthalene balls or thread may shift, changing exposure to airflow.
- Airflow and temperature changes may alter mass transfer rates.

Industrial relevance:

- **Chemical Processing** – Sublimation in fine chemical and intermediate production.
- **Pharmaceuticals** – Drying, freeze-drying, and sublimation in drug formulation.
- **Food Industry** – Freeze-drying of food products to enhance shelf life.
- **Air Fresheners & Coatings** – Controlled release of fragrances and volatile compounds.
- **Pollution Control** – Gas purification using solid sorbents in air filtration.
- **Catalysis & Energy** – Fluidized beds for efficient solid-gas reactions in reactors.
- **Textile & Paper Industry** – Drying and solvent recovery during manufacturing.

Academic relevance:

The study of solid-gas mass transfer, as demonstrated in this experiment, holds significant academic relevance in chemical, environmental, and process engineering. It provides a

fundamental understanding of diffusion, adsorption, and sublimation—key principles in designing efficient industrial processes. These concepts are integral to advanced research in areas such as reactor design, air pollution control, and gas separation technologies.

Furthermore, the experiment reinforces theoretical knowledge by offering practical insights into mass transfer coefficients, diffusion rates, and concentration gradients. This hands-on approach bridges the gap between fundamental principles and real-world applications, preparing students for future research in simulation modeling, material science, and sustainable engineering solutions.

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